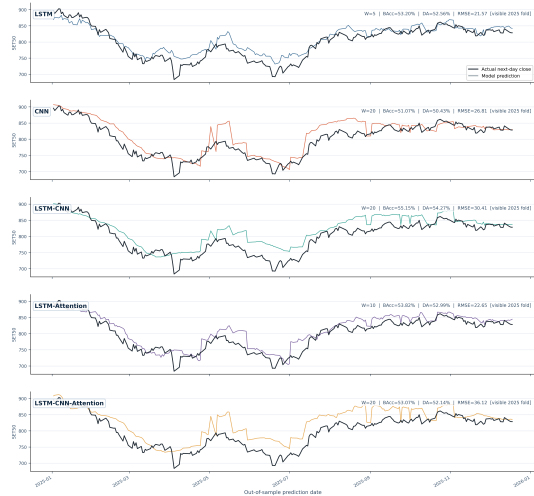
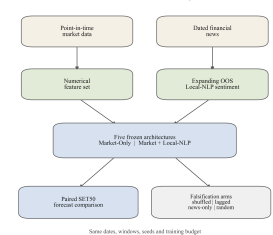


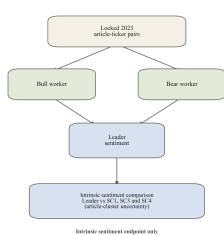


SET50 forecasts (2025 OOS)

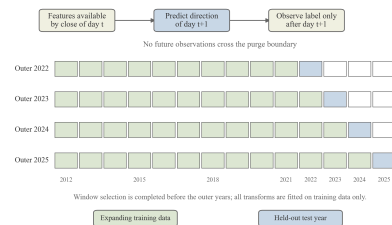
A Downstream SET50 forecasting audit



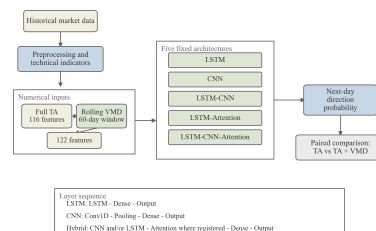
B Locked intrinsic LLM sentiment audit



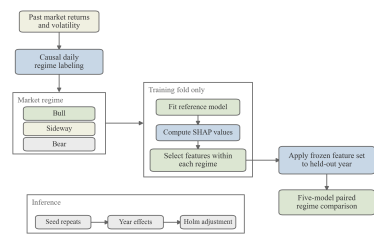
Dated news + Bull/Bear/Leader



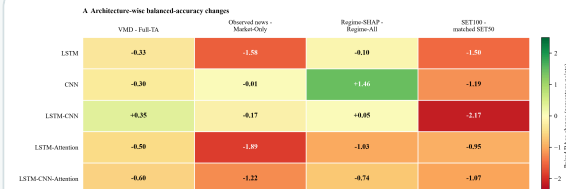
Point-in-time expanding folds



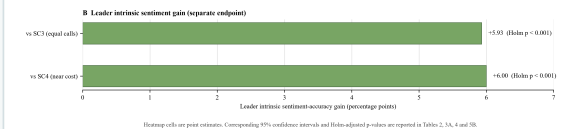
Causal VMD + five frozen models



Causal regime routing + XAI

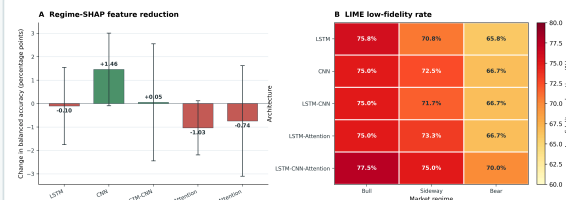


Green indicates a positive point estimate; red indicates a negative point estimate. No holm-adjusted correction (false adjustment).



Hotmap cells are point estimates. Corresponding 95% confidence intervals and Holm-adjusted p-values are reported in Tables 2, 3A, 4 and 5B.

Cross-track effects



SHAP/LIME diagnostics

Final five-model results on the integrated out-of-sample arm

Regime-SHAP-Numerical-News | mean across four temporal folds (2022-2023). Five seeds per fold

Rank	Model	Window	Balanced Accuracy (%)	Direction Accuracy (%)	BCC	RMSE	MAE	Temporal Folds
1	LSTM-CNN-Attention	20	53.84 ± 0.42	53.43 ± 0.34	0.0003	20.73	20.24	4
2	LSTM-CNN	20	52.81 ± 2.29	52.13 ± 2.58	0.0057	29.79	23.44	4
3	LSTM-Attention	10	52.62 ± 1.46	52.40 ± 0.80	0.0018	20.41	13.90	4
4	LSTM	5	52.03 ± 1.48	51.98 ± 0.43	0.0443	15.97	12.73	4
5	CNN	10	51.49 ± 1.83	51.76 ± 1.75	0.0308	22.74	17.73	4

Primary ranking metric: balanced accuracy. Lower RMSE/MAE is better. Higher balanced accuracy, direction accuracy, and BCC is better.

Final five-model held-out panel